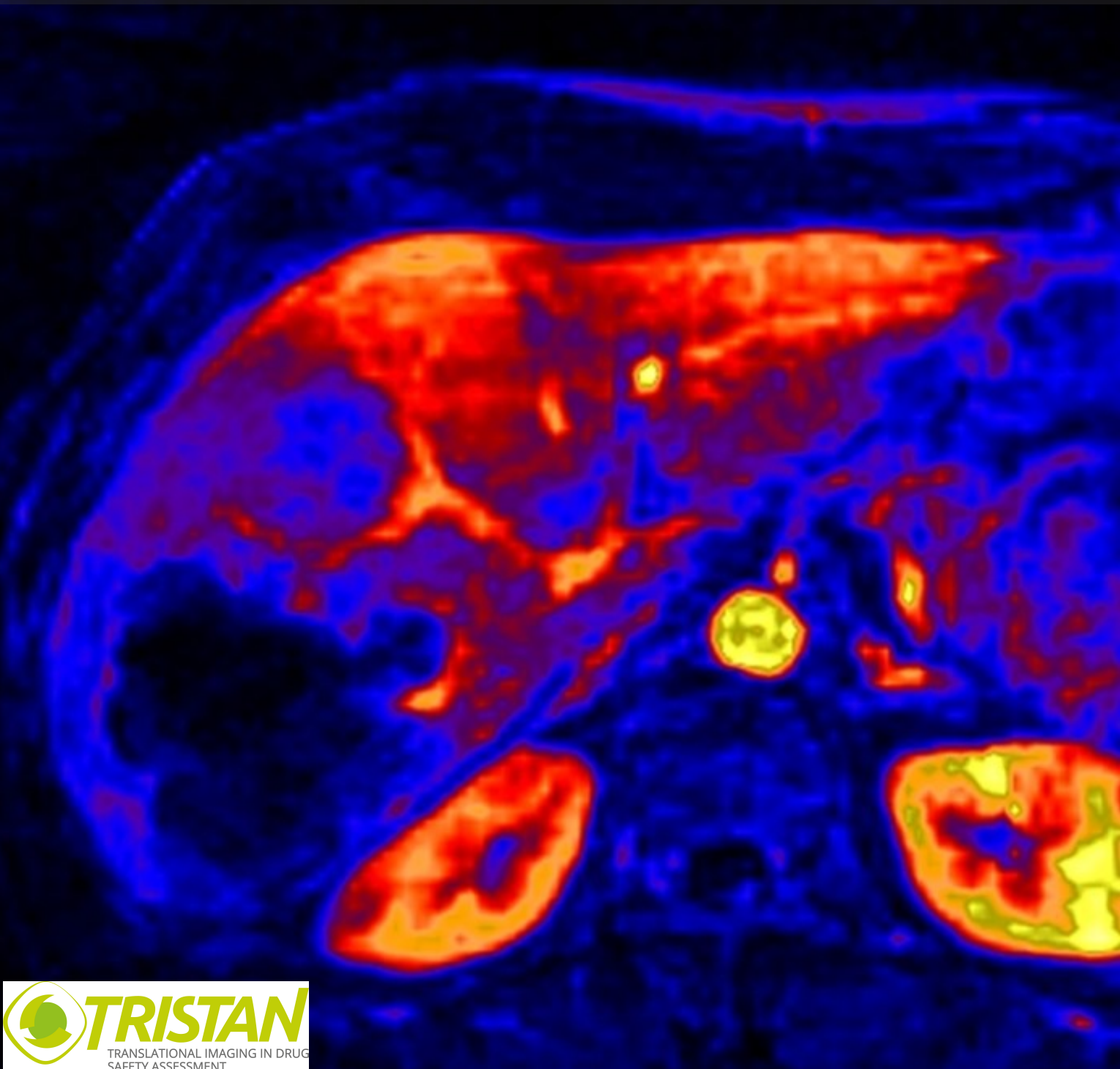


Leeds pilot study

rifampicin (all results)

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by

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Two-scan results

1.1. Data summary

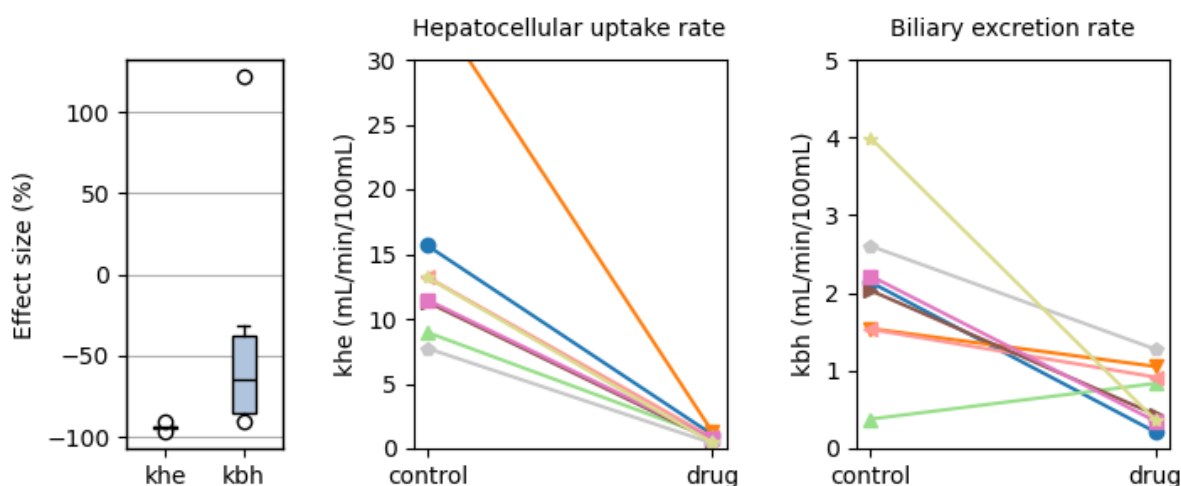


Figure 1.1: Effect size (%) on hepatocellular uptake (k_{he} , left) and biliary excretion (k_{bh} , right) of gadoxetate. The boxplot shows median, interquartile range and 95 percent range. The line plots show individual values for hepatocellular uptake (k_{he} , middle) and biliary excretion (k_{bh} , right) of gadoxetate of the control (left of plot) and treatment (right of plot). Grey lines are healthy controls with rifampicin injection.

	parameter	count	mean	std	min	25%	50%	75%	max
	khe effect size (%)	8.0	-94.0	1.9	-96.3	-95.0	-93.9	-93.2	-90.5
	khe control (mL/min/100cm ³)	8.0	14.55	8.59	7.72	10.71	12.32	13.83	34.88
	khe drug (mL/min/100cm ³)	8.0	0.81	0.31	0.42	0.61	0.8	0.91	1.34
	kbh effect size (%)	8.0	-43.1	70.4	-90.8	-85.6	-64.7	-38.0	121.3
	kbh control (mL/min/100cm ³)	8.0	2.05	1.03	0.38	1.54	2.08	2.31	3.99
	kbh drug (mL/min/100cm ³)	8.0	0.68	0.39	0.21	0.36	0.64	0.95	1.28

Table 1.1: Effect size and absolute values of hepatocellular uptake (k_{he}) and biliary excretion (k_{bh}) of gadoxetate

1.2. Liver biomarkers

Biomarker	p-value	Bayes Factor	Odds Ratio
AUC for CI (0-35min)	0.0001	264.17	1017.19
AUC for CI (0-inf)	0.00199	23.64	50.25
Biliary excretion rate	0.01533	4.68	24.4
Biliary tissue excretion rate	0.01661	4.4	20.69
Final biliary excretion rate	0.07399	1.42	6.03
Final hepatocellular uptake rate	0.02396	3.31	13.0
Gut dispersion	0.37833	0.48	0.42
Gut mean transit time	0.30326	0.54	0.32
Hepatocellular mean transit time	0.16657	0.8	0.21
Hepatocellular tissue uptake rate	0.02861	2.89	12.13
Hepatocellular uptake rate	0.00232	20.86	60.47
Initial biliary excretion rate	0.12207	0.99	4.95
Initial hepatocellular uptake rate	2e-05	1054.0	9919.24
Liver T1-MOLLI at 45min	0.0	6244.03	0.0
Liver T1-MOLLI at baseline	0.3834	0.48	0.73
Liver T1-MOLLI at scan 2	0.00215	22.19	0.06
Liver extracellular volume fraction	0.03628	2.42	0.1
Liver plasma clearance	0.00187	24.85	73.73
RE for R11 at 20min	7e-05	376.64	1013.36
RE for SI at 20min	8e-05	342.72	813.84

Table 1.2: Results of a pairwise comparison testing for differences in liver biomarkers between control and treatment. The results are ranked by their p-value, with most significant differences at the top of the list.

Biomarker	Units	control	drug	change (%)
AUC for CI (0-35min)	mM*sec	177.0 (36.0)	32.8 (5.8)	-80.4 (4.1)
AUC for CI (0-inf)	mM*sec	562.0 (190.0)	129.0 (25.0)	-73.9 (5.8)
Biliary excretion rate	mL/min/100cm3	2.05 (0.71)	0.682 (0.27)	-43.1 (49.0)
Biliary tissue excretion rate	/min	0.0272 (0.0077)	0.0116 (0.005)	-26.3 (70.0)
Final biliary excretion rate	mL/min/100cm3	1.96 (0.79)	1.04 (0.46)	-13.9 (61.0)
Final hepatocellular uptake rate	mL/min/100cm3	18.4 (12.0)	0.962 (0.26)	-93.0 (2.3)
Gut dispersion	%	61.5 (15.0)	69.3 (4.6)	28.7 (40.0)
Gut mean transit time	sec	33.8 (13.0)	43.3 (7.5)	72.3 (72.0)
Hepatocellular mean transit time	min	60.0 (51.0)	131.0 (64.0)	270.0 (200.0)
Hepatocellular tissue uptake rate	/min	1.77 (1.2)	0.0398 (0.012)	-96.3 (1.4)
Hepatocellular uptake rate	mL/min/100cm3	14.5 (6.0)	0.806 (0.21)	-94.0 (1.3)
Initial biliary excretion rate	mL/min/100cm3	2.62 (0.97)	1.27 (1.1)	-36.7 (52.0)
Initial hepatocellular uptake rate	mL/min/100cm3	10.7 (1.9)	0.649 (0.2)	-93.6 (2.3)
Liver T1-MOLLI at 45min	sec	0.537 (0.036)	0.727 (0.041)	35.7 (5.9)
Liver T1-MOLLI at baseline	sec	0.795 (0.037)	0.805 (0.041)	1.27 (2.7)
Liver T1-MOLLI at scan 2	sec	0.669 (0.041)	0.759 (0.038)	13.8 (6.3)
Liver extracellular volume fraction	mL/100cm3	13.1 (5.4)	21.1 (2.9)	196.0 (230.0)
Liver plasma clearance	L/min	0.13 (0.051)	0.00738 (0.0015)	-93.7 (1.6)
RE for R11 at 20min	%	63.7 (14.0)	8.8 (1.5)	-85.7 (1.8)
RE for SI at 20min	%	53.4 (12.0)	7.83 (1.8)	-84.9 (2.5)

Table 1.3: Mean values along with their 95 percent confidence intervals for all liver biomarkers of the control and treatment visit. The last column shows the relative change at the treatment visit. The results are ranked by their p-value, with most significant differences at the top of the list.

1.3. Systemic biomarkers

Biomarker	p-value	Bayes Factor	Odds Ratio
AUC for Cb (0-35min)	0.05248	1.83	0.34
AUC for Cb (0-inf)	0.05911	1.68	0.22
Body extraction fraction	0.04209	2.16	7.85
Cardiac output	0.2363	0.64	0.4
Heart-lung dispersion	0.07318	1.43	0.18
Heart-lung mean transit time	0.86741	0.34	1.14
Organs blood mean transit time	0.49445	0.42	0.66
Organs extraction fraction	0.00036	94.79	15.81
Organs extravascular mean transit time	0.25136	0.61	0.43
RE for R1b at 20min	0.0241	3.3	0.22
RE for Sb at 20min	0.71646	0.36	0.7

Table 1.4: Results of a pairwise comparison testing for differences in systemic biomarkers between control and treatment visit. The results are ranked by their p-value, with most significant differences at the top of the list.

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	45.3 (7.3)	54.0 (12.0)	17.2 (16.0)
AUC for Cb (0-inf)	mM*sec	74.8 (29.0)	110.0 (30.0)	59.5 (42.0)
Body extraction fraction	%	6.81 (2.8)	3.49 (0.67)	-32.1 (29.0)
Cardiac output	L/min	5.07 (1.4)	5.89 (0.8)	29.2 (30.0)
Heart-lung dispersion	%	37.5 (6.2)	46.9 (7.3)	32.0 (31.0)
Heart-lung mean transit time	sec	17.5 (3.2)	17.2 (2.6)	3.48 (21.0)
Organs blood mean transit time	sec	31.1 (8.1)	33.6 (7.2)	15.3 (22.0)
Organs extraction fraction	%	20.4 (3.0)	14.2 (2.6)	-29.9 (7.8)
Organs extravascular mean transit time	min	6.35 (2.3)	7.56 (1.1)	44.6 (50.0)
RE for R1b at 20min	%	17.1 (3.2)	22.7 (5.8)	30.5 (23.0)
RE for Sb at 20min	%	16.2 (6.0)	18.1 (6.6)	40.6 (64.0)

Table 1.5: Mean values along with their 95 percent confidence intervals for all systemic biomarkers at the control and treatment visit. The last column shows the relative change at the treatment visit. The results are ranked by their p-value, with most significant differences at the top of the list.

1.4. Case notes
1.4.1. Subject LDS-002

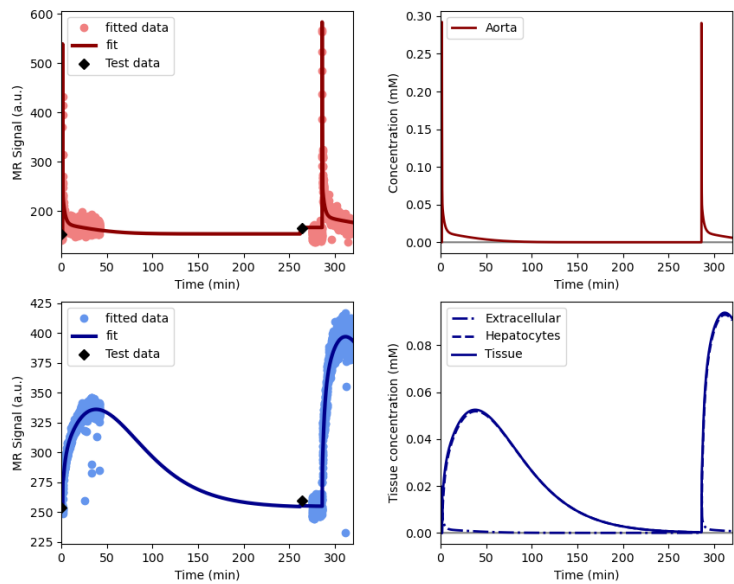


Figure 1.2: Signal-time curves for subject LDS-002 at the control visit.

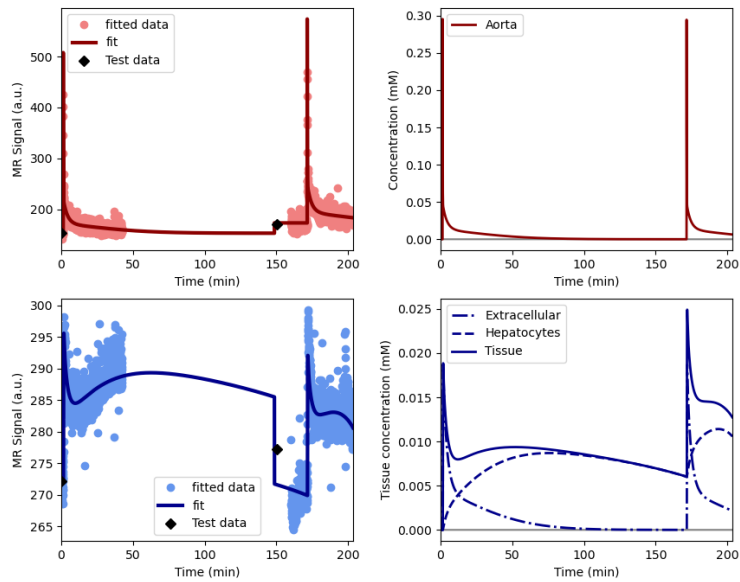


Figure 1.3: Signal-time curves for subject LDS-002 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for CI (0-35min)	mM*sec	107.3	19.11	-82.19
AUC for CI (0-inf)	mM*sec	348.95	93.94	-73.08
Biliary excretion rate	mL/min/100cm3	2.14	0.21	-90.06
Biliary tissue excretion rate	/min	0.02	0.0	-86.84
Final biliary excretion rate	mL/min/100cm3	2.9	2.18	-24.79
Final hepatocellular uptake rate	mL/min/100cm3	22.31	1.29	-94.2
Gut dispersion	%	49.33	69.96	41.83
Gut mean transit time	sec	15.59	35.82	129.81
Hepatocellular mean transit time	min	41.22	313.19	659.75
Hepatocellular tissue uptake rate	/min	2.39	0.06	-97.51
Hepatocellular uptake rate	mL/min/100cm3	15.65	1.09	-93.01
Initial biliary excretion rate	mL/min/100cm3	1.69	0.11	-93.41
Initial hepatocellular uptake rate	mL/min/100cm3	9.0	0.9	-90.04
Liver T1-MOLLI at 45min	sec	0.58	0.72	23.26
Liver T1-MOLLI at baseline	sec	0.76	0.81	7.72
Liver T1-MOLLI at scan 2	sec	0.74	0.75	1.88
Liver extracellular volume fraction	mL/100cm3	6.55	18.42	181.28
Liver plasma clearance	L/min	0.11	0.01	-91.79
RE for R1l at 20min	%	35.95	5.94	-83.47
RE for Sl at 20min	%	27.31	4.9	-82.06

Table 1.6: Values for liver of subject LDS-002

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	26.88	26.15	-2.72
AUC for Cb (0-inf)	mM*sec	40.59	39.14	-3.56
Body extraction fraction	%	8.4	5.27	-37.29
Cardiac output	L/min	2.79	5.25	88.18
Heart-lung dispersion	%	52.23	46.9	-10.2
Heart-lung mean transit time	sec	23.76	15.45	-34.96
Organs blood mean transit time	sec	36.61	28.58	-21.94
Organs extraction fraction	%	22.68	13.51	-40.45
Organs extravascular mean transit time	min	7.39	7.7	4.31
RE for R1b at 20min	%	9.54	8.89	-6.8
RE for Sb at 20min	%	9.33	7.71	-17.36

Table 1.7: Values for aorta of subject LDS-002

1.4.2. Subject LDS-003

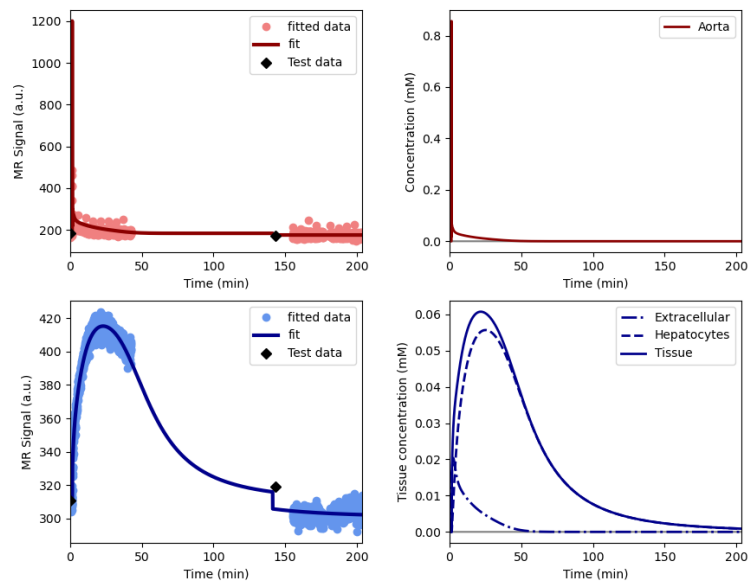


Figure 1.4: Signal-time curves for subject LDS-003 at the control visit.

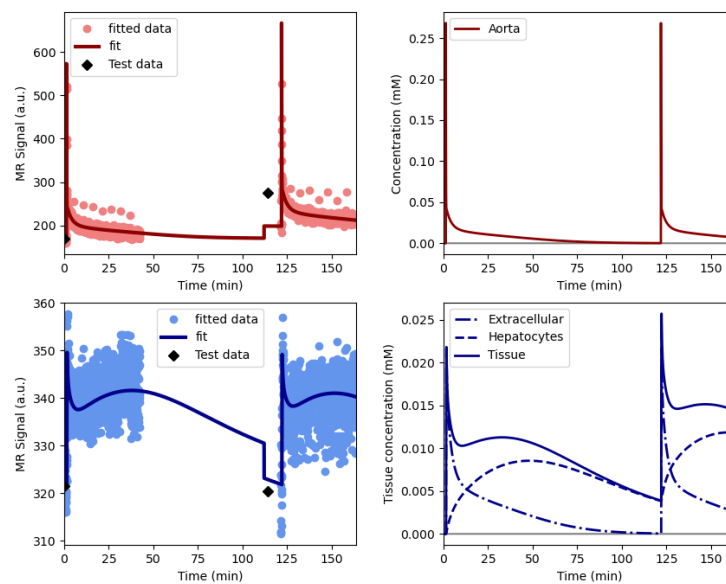


Figure 1.5: Signal-time curves for subject LDS-003 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for Cl (0-35min)	mM*sec	270.83	23.66	-91.26
AUC for Cl (0-inf)	mM*sec	242.47	78.35	-67.69
Biliary excretion rate	mL/min/100cm3	1.54	1.05	-31.54
Biliary tissue excretion rate	/min	0.03	0.02	-39.02
Final biliary excretion rate	mL/min/100cm3	0.92	1.14	24.56
Final hepatocellular uptake rate	mL/min/100cm3	60.0	1.53	-97.45
Gut dispersion	%	87.68	64.27	-26.69
Gut mean transit time	sec	58.89	28.2	-52.12
Hepatocellular mean transit time	min	36.77	60.29	63.99
Hepatocellular tissue uptake rate	/min	1.46	0.07	-95.41
Hepatocellular uptake rate	mL/min/100cm3	34.88	1.34	-96.15
Initial biliary excretion rate	mL/min/100cm3	4.79	0.98	-79.56
Initial hepatocellular uptake rate	mL/min/100cm3	9.75	1.16	-88.13
Liver T1-MOLLI at 45min	sec	0.53	0.69	30.25
Liver T1-MOLLI at baseline	sec	0.71	0.73	2.64
Liver T1-MOLLI at scan 2	sec	0.66	0.71	7.58
Liver extracellular volume fraction	mL/100cm3	23.85	20.02	-16.03
Liver plasma clearance	L/min	0.3	0.01	-96.2
RE for R1l at 20min	%	42.4	6.72	-84.15
RE for Sl at 20min	%	36.02	4.63	-87.14

Table 1.8: Values for liver of subject LDS-003

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	36.32	30.72	-15.4
AUC for Cb (0-inf)	mM*sec	39.92	49.33	23.57
Body extraction fraction	%	5.02	3.49	-30.5
Cardiac output	L/min	6.07	7.06	16.3
Heart-lung dispersion	%	20.18	43.36	114.86
Heart-lung mean transit time	sec	9.96	15.05	51.15
Organs blood mean transit time	sec	20.04	24.77	23.62
Organs extraction fraction	%	15.04	11.55	-23.19
Organs extravascular mean transit time	min	2.02	6.0	197.39
RE for R1b at 20min	%	13.31	12.66	-4.93
RE for Sb at 20min	%	28.87	4.16	-85.59

Table 1.9: Values for aorta of subject LDS-003

1.4.3. Subject LDS-004

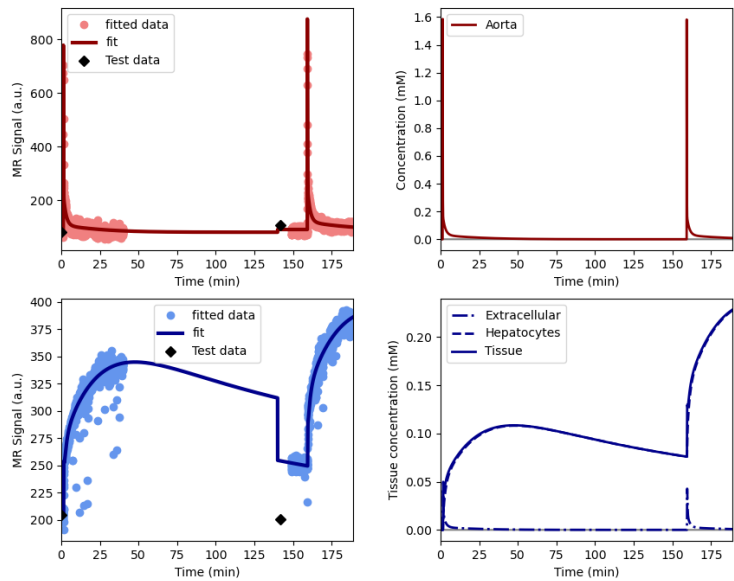


Figure 1.6: Signal-time curves for subject LDS-004 at the control visit.

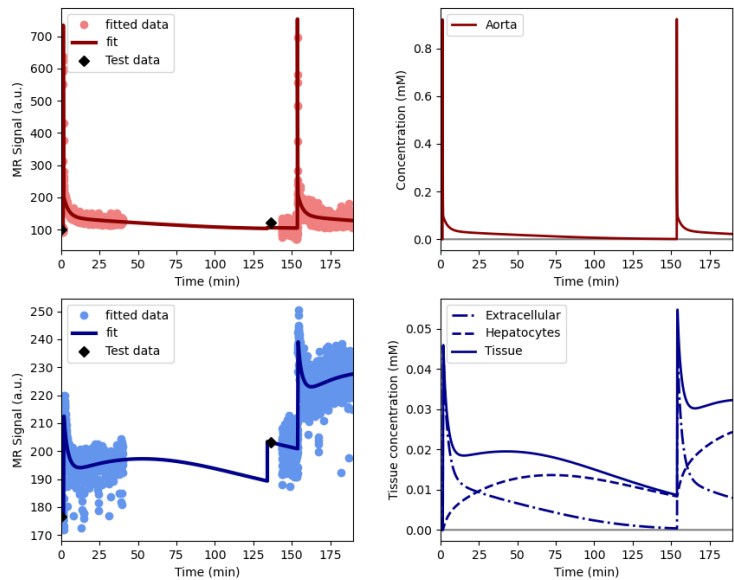


Figure 1.7: Signal-time curves for subject LDS-004 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for CI (0-35min)	mM*sec	196.84	45.11	-77.09
AUC for CI (0-inf)	mM*sec	1037.68	176.14	-83.03
Biliary excretion rate	mL/min/100cm3	0.38	0.84	121.3
Biliary tissue excretion rate	/min	0.0	0.01	213.87
Final biliary excretion rate	mL/min/100cm3	0.27	0.76	183.2
Final hepatocellular uptake rate	mL/min/100cm3	10.84	1.08	-90.07
Gut dispersion	%	50.35	80.31	59.52
Gut mean transit time	sec	23.02	52.72	129.02
Hepatocellular mean transit time	min	241.09	76.81	-68.14
Hepatocellular tissue uptake rate	/min	1.82	0.04	-97.63
Hepatocellular uptake rate	mL/min/100cm3	8.95	0.85	-90.49
Initial biliary excretion rate	mL/min/100cm3	0.64	0.93	45.79
Initial hepatocellular uptake rate	mL/min/100cm3	7.06	0.63	-91.15
Liver T1-MOLLI at 45min	sec	0.5	0.71	41.3
Liver T1-MOLLI at baseline	sec	0.8	0.8	-0.01
Liver T1-MOLLI at scan 2	sec	0.65	0.74	14.22
Liver extracellular volume fraction	mL/100cm3	4.91	19.68	300.84
Liver plasma clearance	L/min	0.08	0.01	-89.45
RE for R1l at 20min	%	75.11	12.07	-83.94
RE for Sl at 20min	%	60.17	12.61	-79.04

Table 1.10: Values for liver of subject LDS-004

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	55.47	74.89	35.02
AUC for Cb (0-inf)	mM*sec	68.13	152.91	124.43
Body extraction fraction	%	15.0	4.4	-70.7
Cardiac output	L/min	2.68	3.88	44.97
Heart-lung dispersion	%	39.77	39.54	-0.59
Heart-lung mean transit time	sec	14.83	18.23	22.96
Organs blood mean transit time	sec	34.69	47.62	37.25
Organs extraction fraction	%	27.78	18.52	-33.35
Organs extravascular mean transit time	min	6.89	7.83	13.67
RE for R1b at 20min	%	17.16	31.17	81.71
RE for Sb at 20min	%	30.13	24.4	-19.02

Table 1.11: Values for aorta of subject LDS-004

1.4.4. Subject LDS-006

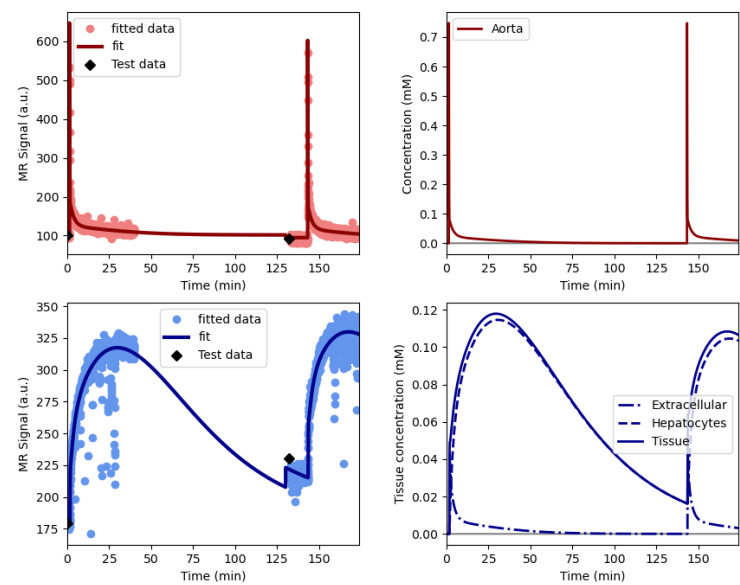


Figure 1.8: Signal-time curves for subject LDS-006 at the control visit.

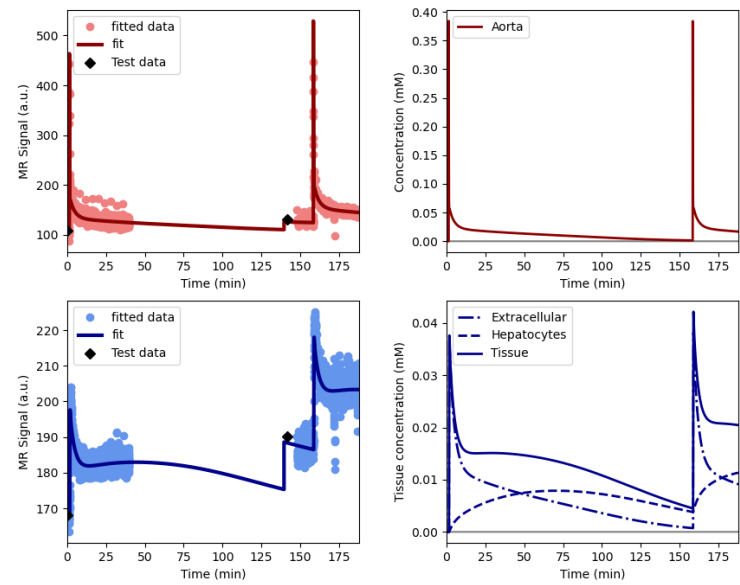


Figure 1.9: Signal-time curves for subject LDS-006 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for Cl (0-35min)	mM*sec	202.43	35.6	-82.42
AUC for Cl (0-inf)	mM*sec	662.2	125.43	-81.06
Biliary excretion rate	mL/min/100cm3	1.53	0.92	-40.09
Biliary tissue excretion rate	/min	0.02	0.02	-12.56
Final biliary excretion rate	mL/min/100cm3	1.62	1.11	-31.69
Final hepatocellular uptake rate	mL/min/100cm3	12.06	1.08	-91.05
Gut dispersion	%	62.3	64.81	4.02
Gut mean transit time	sec	43.81	41.65	-4.92
Hepatocellular mean transit time	min	43.96	50.28	14.37
Hepatocellular tissue uptake rate	/min	0.74	0.03	-96.12
Hepatocellular uptake rate	mL/min/100cm3	13.22	0.84	-93.61
Initial biliary excretion rate	mL/min/100cm3	1.45	0.78	-46.04
Initial hepatocellular uptake rate	mL/min/100cm3	14.37	0.61	-95.76
Liver T1-MOLLI at 45min	sec	0.5	0.75	51.37
Liver T1-MOLLI at baseline	sec	0.84	0.84	-0.39
Liver T1-MOLLI at scan 2	sec	0.68	0.79	15.44
Liver extracellular volume fraction	mL/100cm3	17.96	29.62	64.91
Liver plasma clearance	L/min	0.09	0.01	-93.47
RE for R1l at 20min	%	92.96	9.53	-89.74
RE for Sl at 20min	%	74.66	7.95	-89.35

Table 1.12: Values for liver of subject LDS-006

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	42.18	47.1	11.68
AUC for Cb (0-inf)	mM*sec	58.21	110.96	90.62
Body extraction fraction	%	8.75	2.27	-74.0
Cardiac output	L/min	3.74	6.96	85.96
Heart-lung dispersion	%	37.38	39.54	5.78
Heart-lung mean transit time	sec	17.81	17.82	0.07
Organs blood mean transit time	sec	36.18	28.01	-22.58
Organs extraction fraction	%	20.61	12.91	-37.35
Organs extravascular mean transit time	min	6.91	7.67	10.94
RE for R1b at 20min	%	15.25	20.82	36.55
RE for Sb at 20min	%	6.83	20.32	197.32

Table 1.13: Values for aorta of subject LDS-006

1.4.5. Subject LDS-007

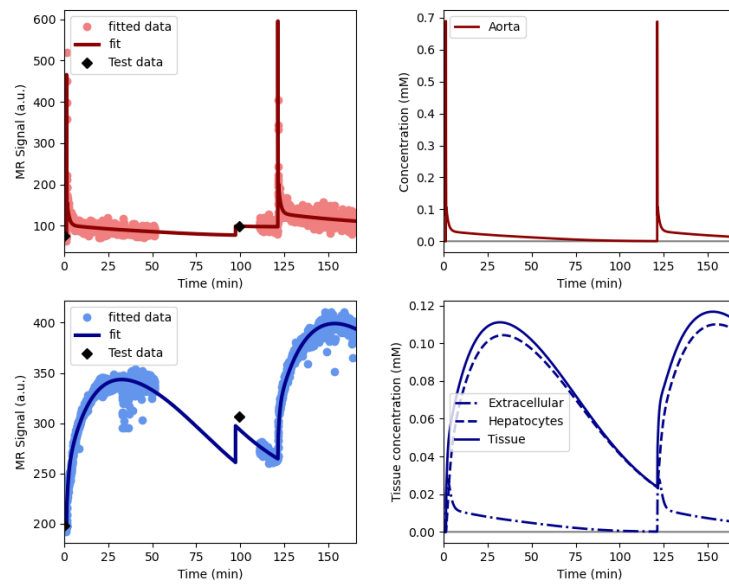


Figure 1.10: Signal-time curves for subject LDS-007 at the control visit.

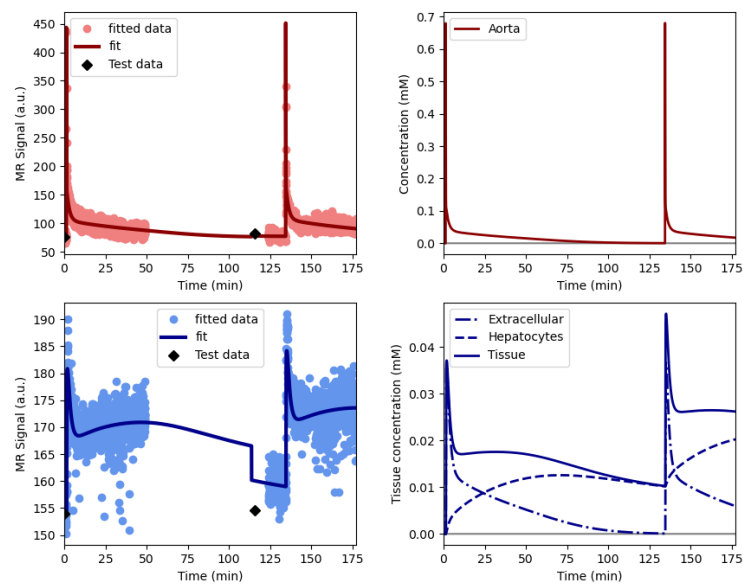


Figure 1.11: Signal-time curves for subject LDS-007 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for Cl (0-35min)	mM*sec	192.02	40.11	-79.11
AUC for Cl (0-inf)	mM*sec	640.43	159.76	-75.05
Biliary excretion rate	mL/min/100cm3	2.03	0.44	-78.54
Biliary tissue excretion rate	/min	0.03	0.01	-79.97
Final biliary excretion rate	mL/min/100cm3	1.71	0.23	-86.7
Final hepatocellular uptake rate	mL/min/100cm3	9.83	0.49	-95.0
Gut dispersion	%	95.71	74.07	-22.61
Gut mean transit time	sec	60.0	41.94	-30.09
Hepatocellular mean transit time	min	30.34	151.47	399.16
Hepatocellular tissue uptake rate	/min	0.54	0.03	-93.52
Hepatocellular uptake rate	mL/min/100cm3	11.3	0.65	-94.26
Initial biliary excretion rate	mL/min/100cm3	2.5	5.25	110.16
Initial hepatocellular uptake rate	mL/min/100cm3	12.77	0.81	-93.69
Liver T1-MOLLI at 45min	sec	0.54	0.75	40.39
Liver T1-MOLLI at baseline	sec	0.86	0.84	-1.75
Liver T1-MOLLI at scan 2	sec	0.6	0.8	33.05
Liver extracellular volume fraction	mL/100cm3	21.11	18.7	-11.42
Liver plasma clearance	L/min	0.11	0.01	-93.22
RE for R1l at 20min	%	85.87	11.67	-86.41
RE for Sl at 20min	%	75.07	9.46	-87.39

Table 1.14: Values for liver of subject LDS-007

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	55.92	68.5	22.49
AUC for Cb (0-inf)	mM*sec	97.73	121.98	24.81
Body extraction fraction	%	3.52	3.53	0.2
Cardiac output	L/min	6.66	5.64	-15.31
Heart-lung dispersion	%	32.54	42.61	30.95
Heart-lung mean transit time	sec	13.98	14.0	0.11
Organs blood mean transit time	sec	14.31	21.08	47.26
Organs extraction fraction	%	23.99	20.34	-15.21
Organs extravascular mean transit time	min	3.78	4.38	15.75
RE for R1b at 20min	%	24.96	29.21	17.02
RE for Sb at 20min	%	14.43	34.27	137.39

Table 1.15: Values for aorta of subject LDS-007

1.4.6. Subject LDS-008

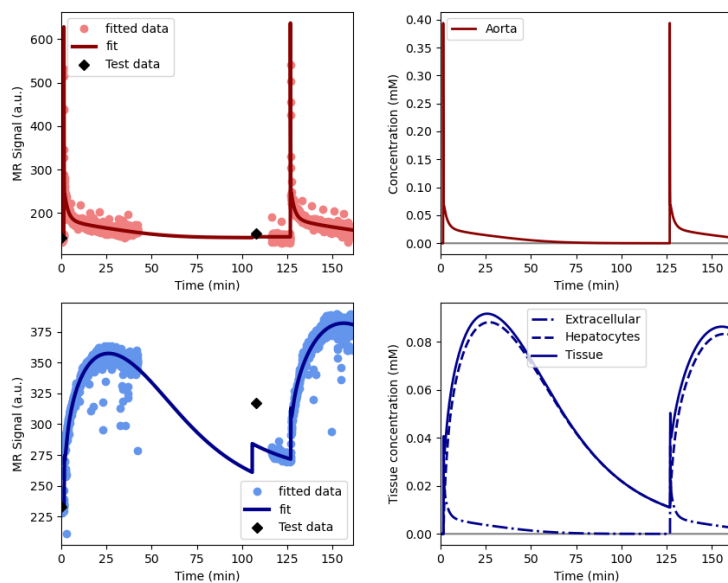


Figure 1.12: Signal-time curves for subject LDS-008 at the control visit.

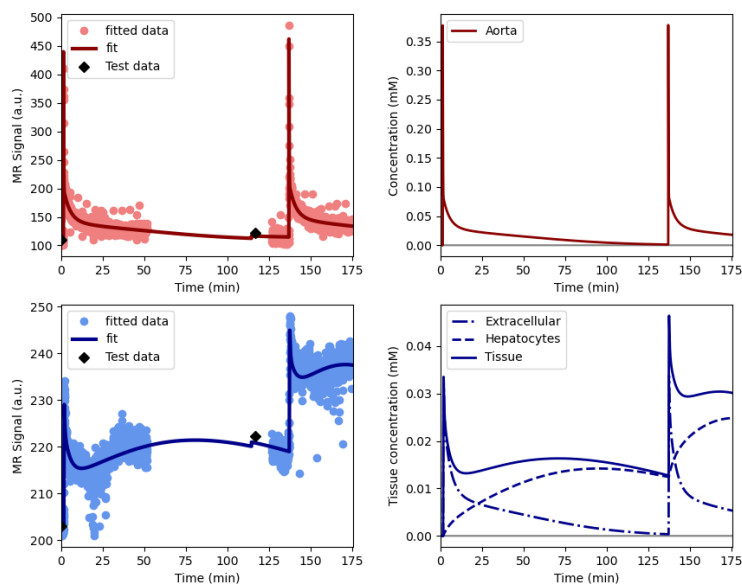


Figure 1.13: Signal-time curves for subject LDS-008 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for Cl (0-35min)	mM*sec	165.96	33.1	-80.06
AUC for Cl (0-inf)	mM*sec	441.39	164.13	-62.81
Biliary excretion rate	mL/min/100cm3	2.22	0.35	-84.15
Biliary tissue excretion rate	/min	0.03	0.0	-83.24
Final biliary excretion rate	mL/min/100cm3	1.69	0.99	-40.98
Final hepatocellular uptake rate	mL/min/100cm3	8.75	1.08	-87.68
Gut dispersion	%	49.98	58.82	17.7
Gut mean transit time	sec	19.3	33.68	74.52
Hepatocellular mean transit time	min	33.64	200.71	496.7
Hepatocellular tissue uptake rate	/min	0.82	0.05	-94.24
Hepatocellular uptake rate	mL/min/100cm3	11.46	0.76	-93.33
Initial biliary excretion rate	mL/min/100cm3	3.24	0.21	-93.42
Initial hepatocellular uptake rate	mL/min/100cm3	14.17	0.45	-96.81
Liver T1-MOLLI at 45min	sec	0.46	0.62	33.65
Liver T1-MOLLI at baseline	sec	0.74	0.71	-5.07
Liver T1-MOLLI at scan 2	sec	0.57	0.66	14.25
Liver extracellular volume fraction	mL/100cm3	13.97	16.18	15.85
Liver plasma clearance	L/min	0.12	0.01	-94.29
RE for R1l at 20min	%	65.04	7.64	-88.26
RE for Sl at 20min	%	52.32	7.0	-86.63

Table 1.16: Values for liver of subject LDS-008

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	43.43	63.55	46.31
AUC for Cb (0-inf)	mM*sec	61.22	125.98	105.78
Body extraction fraction	%	4.91	3.33	-32.3
Cardiac output	L/min	8.09	5.66	-30.08
Heart-lung dispersion	%	41.21	49.01	18.94
Heart-lung mean transit time	sec	17.97	23.63	31.54
Organs blood mean transit time	sec	22.48	29.92	33.07
Organs extraction fraction	%	16.23	11.55	-28.82
Organs extravascular mean transit time	min	4.61	9.09	97.17
RE for R1b at 20min	%	18.43	25.21	36.83
RE for Sb at 20min	%	13.54	21.29	57.26

Table 1.17: Values for aorta of subject LDS-008

1.4.7. Subject LDS-009

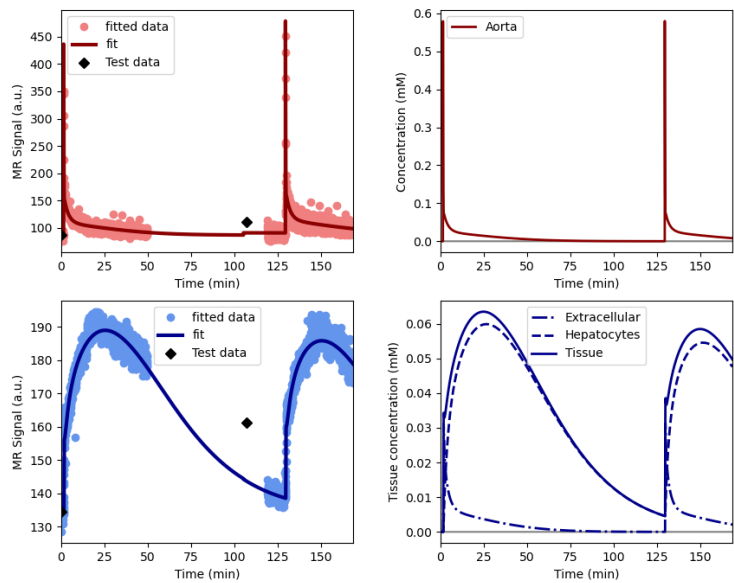


Figure 1.14: Signal-time curves for subject LDS-009 at the control visit.

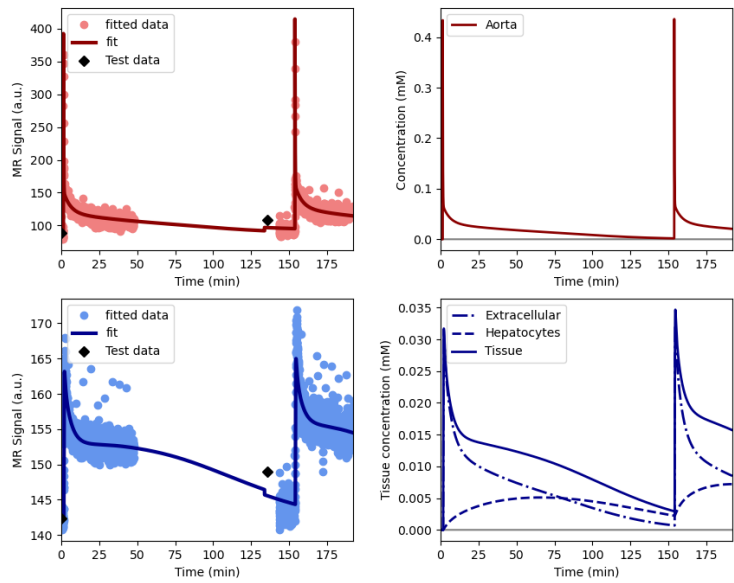


Figure 1.15: Signal-time curves for subject LDS-009 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for Cl (0-35min)	mM*sec	112.25	33.4	-70.25
AUC for Cl (0-inf)	mM*sec	288.12	101.56	-64.75
Biliary excretion rate	mL/min/100cm3	2.6	1.28	-50.95
Biliary tissue excretion rate	/min	0.03	0.02	-38.14
Final biliary excretion rate	mL/min/100cm3	2.81	1.65	-41.36
Final hepatocellular uptake rate	mL/min/100cm3	7.33	0.54	-92.69
Gut dispersion	%	67.57	69.15	2.34
Gut mean transit time	sec	34.51	58.11	68.39
Hepatocellular mean transit time	min	28.61	46.25	61.66
Hepatocellular tissue uptake rate	/min	0.55	0.02	-96.63
Hepatocellular uptake rate	mL/min/100cm3	7.72	0.42	-94.6
Initial biliary excretion rate	mL/min/100cm3	2.43	1.04	-57.02
Initial hepatocellular uptake rate	mL/min/100cm3	8.12	0.3	-96.32
Liver T1-MOLLI at 45min	sec	0.62	0.82	31.46
Liver T1-MOLLI at baseline	sec	0.85	0.88	4.05
Liver T1-MOLLI at scan 2	sec	0.73	0.83	14.19
Liver extracellular volume fraction	mL/100cm3	14.04	22.52	60.42
Liver plasma clearance	L/min	0.09	0.01	-94.53
RE for R1l at 20min	%	51.13	8.96	-82.48
RE for Sl at 20min	%	43.1	7.88	-81.72

Table 1.18: Values for liver of subject LDS-009

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	46.16	65.46	41.82
AUC for Cb (0-inf)	mM*sec	63.67	148.19	132.73
Body extraction fraction	%	6.84	2.49	-63.68
Cardiac output	L/min	6.22	7.28	17.02
Heart-lung dispersion	%	39.08	71.65	83.34
Heart-lung mean transit time	sec	18.32	12.23	-33.23
Organs blood mean transit time	sec	32.67	47.66	45.88
Organs extraction fraction	%	17.61	9.58	-45.63
Organs extravascular mean transit time	min	5.87	8.78	49.68
RE for R1b at 20min	%	18.29	31.23	70.73
RE for Sb at 20min	%	16.15	15.34	-5.05

Table 1.19: Values for aorta of subject LDS-009

1.4.8. Subject LDS-010

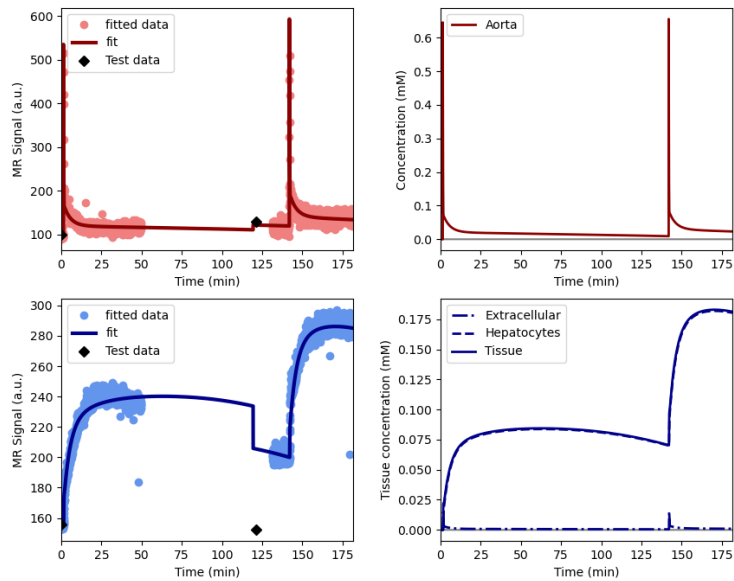


Figure 1.16: Signal-time curves for subject LDS-010 at the control visit.

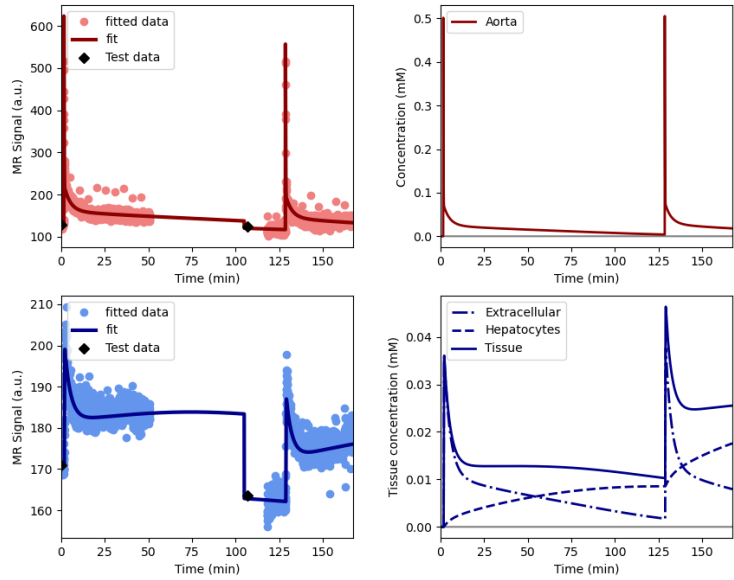


Figure 1.17: Signal-time curves for subject LDS-010 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for Cl (0-35min)	mM*sec	165.15	32.15	-80.53
AUC for Cl (0-inf)	mM*sec	833.41	133.41	-83.99
Biliary excretion rate	mL/min/100cm3	3.99	0.37	-90.76
Biliary tissue excretion rate	/min	0.04	0.01	-84.25
Final biliary excretion rate	mL/min/100cm3	3.79	0.24	-93.75
Final hepatocellular uptake rate	mL/min/100cm3	16.35	0.62	-96.24
Gut dispersion	%	28.74	72.95	153.78
Gut mean transit time	sec	15.0	54.63	264.17
Hepatocellular mean transit time	min	24.03	152.51	534.75
Hepatocellular tissue uptake rate	/min	5.86	0.02	-99.66
Hepatocellular uptake rate	mL/min/100cm3	13.18	0.48	-96.35
Initial biliary excretion rate	mL/min/100cm3	4.21	0.83	-80.24
Initial hepatocellular uptake rate	mL/min/100cm3	10.02	0.35	-96.53
Liver T1-MOLLI at 45min	sec	0.56	0.75	34.2
Liver T1-MOLLI at baseline	sec	0.81	0.83	2.99
Liver T1-MOLLI at scan 2	sec	0.72	0.79	9.82
Liver extracellular volume fraction	mL/100cm3	2.25	24.04	968.58
Liver plasma clearance	L/min	0.14	0.01	-96.24
RE for R1l at 20min	%	61.37	7.85	-87.22
RE for Sl at 20min	%	58.26	8.19	-85.94

Table 1.20: Values for liver of subject LDS-010

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	56.21	55.4	-1.45
AUC for Cb (0-inf)	mM*sec	169.09	130.76	-22.66
Body extraction fraction	%	2.07	3.14	51.37
Cardiac output	L/min	4.28	5.41	26.35
Heart-lung dispersion	%	37.53	42.41	13.01
Heart-lung mean transit time	sec	23.25	20.96	-9.83
Organs blood mean transit time	sec	51.46	41.2	-19.95
Organs extraction fraction	%	18.86	15.95	-15.43
Organs extravascular mean transit time	min	13.33	9.06	-32.07
RE for R1b at 20min	%	19.99	22.65	13.29
RE for Sb at 20min	%	10.67	17.03	59.61

Table 1.21: Values for aorta of subject LDS-010

2

Secondary results

2.1. Diurnal variation

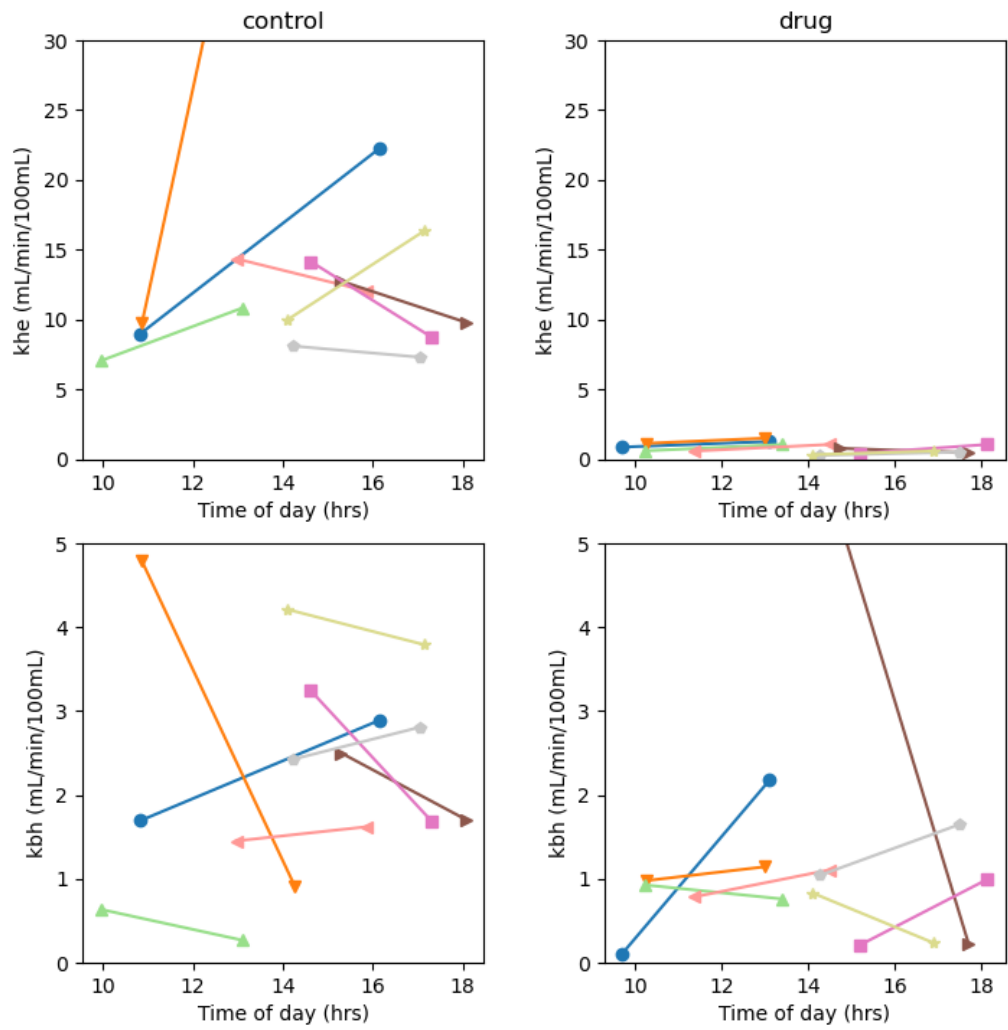


Figure 2.1: Intra-day changes in hepatocellular uptake (k_{he} , top row) and biliary excretion (k_{bh} , bottom row) of gadoxetate at for the control (left column) and treatment (right column). Full lines connect values taken in the same subject at the same day.